

# Chapter 1: Integers

10 worksheets · one for each exercise · Easy, Medium  
and Challenging columns

**NSW syllabus outcome**

Stage 4 Number and algebra: MA4-INT-C-01 compares, orders  
and calculates with integers to solve problems.

NAME \_\_\_\_\_

CLASS \_\_\_\_\_

CHAPTER

1

**1.01** Integers and opposites

**1.02** The number line

**1.03** Comparing and ordering integers

**1.04** Adding integers

**1.05** Subtracting integers

**1.06** Multiplying integers

**1.07** Dividing integers

**1.08** Order of operations with integers

**1.09** Integers on a calculator

**1.10** Integer problems

## How to use this booklet

**1** There is **one worksheet for each lesson**. It has three columns that get harder from left to right: **Easy**, then **Medium**, then **Challenging**.

**2** Read the **speech bubble** at the top of each column: it tells you what to do. A new bubble means a new kind of question.

**WE DO** **Top of each column.** Your teacher works the examples on the board. **Copy every step** into the space.

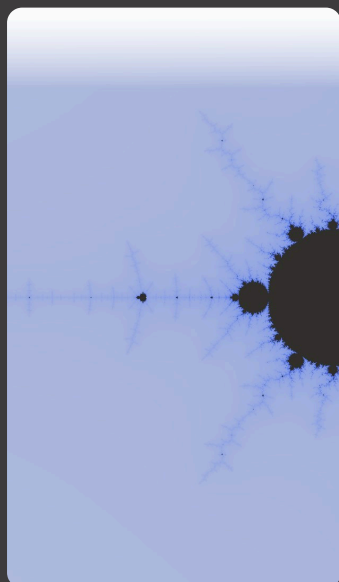
**YOU DO** **The rest of the column.** Your turn. Column 1: answer on the line. Columns 2 and 3: show your working on the lines.

## Contents and progress tracker

| Lesson      | Topic                             | Page      | Date | Done                     |
|-------------|-----------------------------------|-----------|------|--------------------------|
| <b>1.01</b> | Integers and opposites            | <b>3</b>  |      | <input type="checkbox"/> |
| <b>1.02</b> | The number line                   | <b>4</b>  |      | <input type="checkbox"/> |
| <b>1.03</b> | Comparing and ordering integers   | <b>5</b>  |      | <input type="checkbox"/> |
| <b>1.04</b> | Adding integers                   | <b>6</b>  |      | <input type="checkbox"/> |
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| <b>1.06</b> | Multiplying integers              | <b>8</b>  |      | <input type="checkbox"/> |
| <b>1.07</b> | Dividing integers                 | <b>9</b>  |      | <input type="checkbox"/> |
| <b>1.08</b> | Order of operations with integers | <b>10</b> |      | <input type="checkbox"/> |
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## In this chapter you will

- understand positive and negative numbers and their opposites
- use a number line to show and compare integers
- order integers
- add, subtract, multiply and divide integers
- use the order of operations with integers, with and without a calculator
- solve problems involving integers



### About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule,  $z \rightarrow z^2 + c$ , thousands of times for every point. Every booklet shows a different part of the set: this one is the long antenna of the set along the negative real axis, dotted with tiny copies of the whole set. Maths can make beautiful things.

### Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

## Easy

Questions 1–28

Write the opposite of each integer. It is the same distance from 0, on the other side.



**WE DO** With your teacher: copy the working.

E1 16

E2 7

**YOU DO** Your turn.

1 -59

59

2 41

3 9

4 47

5 -25

6 54

7 -14

8 -52

9 -43

10 23

11 27

12 -21

13 -9

14 29

15 -34

16 -2

17 -55

18 57

Write an integer for each situation.



19 63 floors up

20 a fall of 8 points

21 a fall of 48 points

22 50 m below sea level

23 52 minutes late

24 a gain of 4 kg

25 22 m below sea level

26 a loss of \$9

27 48 steps back

28 66 m above sea level

## Medium

Questions 29–38

Simplify. The opposite of the opposite is the number itself.

**WE DO** With your teacher: copy the working.

E1  $-(+26)$

E2  $-(-16)$

**YOU DO** Your turn.

29  $-(+49)$

= the opposite of 49

= -49

30  $-(+57)$

= the opposite of 57

= \_\_\_\_\_

31  $-(-25)$

= the opposite of \_\_\_\_\_

= \_\_\_\_\_

32  $-(-29)$

33  $-(-41)$

34  $-(-50)$

35  $-(-45)$

36  $-(-22)$

37  $-(-32)$

38  $-(-21)$

## Challenging

Questions 39–46

Answer each question. Write your reasoning on the lines.



**WE DO** With your teacher: copy the working.

E1 The opposite of a number is -14. What is the number?

E2 A diver is 7 m below sea level. Write this as an integer, then write its opposite.

**YOU DO** Your turn.

39 List all the integers whose distance from 0 is less than 3.  
from -2 up to 2

40 If  $n = -39$ , find  $-n$  and  $-(-n)$ .

$-n = -(\text{---}) = \text{---}$

$-(-n) = \text{---}$

41 The opposite of a number is -26. What is the number?

42 List all the integers whose distance from 0 is less than 6.

43 If  $n = -3$ , find  $-n$  and  $-(-n)$ .

44 The opposite of a number is -6. What is the number?

45 A diver is 3 m below sea level. Write this as an integer, then write its opposite.

46 List all the integers whose distance from 0 is less than 4.

### Easy

Questions 1–24

Which integer is further to the right on the number line? Write it on the line.



**WE DO** With your teacher: copy the working.

**E1** 5 or -4

**E2** -2 or -3

**YOU DO** Your turn.

**1** 2 or -6

**2**

**2** 8 or -10

**3** 7 or -13

**4** 1 or -2

**5** 14 or -4

**6** -2 or -11

**7** -1 or 11

**8** 14 or 8

**9** -13 or -7

**10** 3 or -12

**11** 10 or 1

**12** 3 or 2

**13** 15 or -5

**14** -11 or 6

**15** -11 or 0

**16** 5 or -15

**17** 5 or -12

**18** -6 or -3

Start at the first number and move along the number line. Where do you land?



**19** start at -1, move 7 right

**20** start at -5, move 4 left

**21** start at 3, move 7 right

**22** start at 5, move 2 left

**23** start at 5, move 4 left

**24** start at 7, move 5 left

### Medium

Questions 25–33

How many units apart are the two integers on the number line?

**WE DO** With your teacher: copy the working.

**E1** -7 and 11

**E2** -8 and 3

**YOU DO** Your turn.

**25** 3 and -1

-1 to 0 is 1, 0 to 3 is 3

1 + 3 = 4

**26** 11 and -12

-12 to 0 is 12, 0 to 11 is 11

\_\_\_\_\_ + \_\_\_\_\_ = \_\_\_\_\_

**27** 12 and -13

\_\_\_\_\_ to \_\_\_\_\_ is \_\_\_\_\_, \_\_\_\_\_ to \_\_\_\_\_

\_\_\_\_\_ + \_\_\_\_\_ = \_\_\_\_\_

**28** 5 and -11

**29** -9 and 11

**30** -14 and 7

Which integer is exactly halfway between?

**31** -10 and -6

**32** -1 and 7

**33** -11 and -3

### Challenging

Questions 34–39

Count the integers or find the halfway point. Show your working.

**WE DO** With your teacher: copy the working.

**E1** How many integers are between -8 and 5?

**E2** How many integers are between -4 and 23?

**YOU DO** Your turn.

**34** Which integer is halfway between -1 and 25?

distance apart: 26

half: 13

\_\_\_\_\_ + \_\_\_\_\_ = \_\_\_\_\_

**35** Which integer is halfway between -13 and 13?

distance apart: \_\_\_\_\_

half: \_\_\_\_\_

\_\_\_\_\_ + \_\_\_\_\_ = \_\_\_\_\_

**36** How many integers are between -30 and 26?

**37** Which integer is halfway between -23 and -1?

**38** How many integers are between -27 and 16?

**39** How many integers are between -13 and 25?

## Easy

Questions 1–28

Write  $<$  or  $>$  in the box. The integer further right is larger.

**WE DO** With your teacher: copy the working.

E1  $-7 \square -8$

E2  $-5 \square -10$

**YOU DO** Your turn.

1  $-1 \square -20$

2  $-12 \square 12$

3  $7 \square -11$

4  $2 \square 10$

5  $11 \square 0$

6  $-9 \square -6$

7  $-4 \square -6$

8  $-1 \square -17$

9  $-5 \square -15$

10  $-10 \square -6$

11  $-10 \square 0$

12  $-19 \square 3$

13  $1 \square 11$

14  $5 \square -10$

15  $-4 \square -2$

16  $9 \square -6$

17  $-11 \square 12$

18  $7 \square -19$

Write the smaller integer.

19  $-7$  or  $-2$

20  $2$  or  $-14$

21  $0$  or  $-14$

22  $7$  or  $-12$

23  $-21$  or  $-1$

24  $6$  or  $-26$

25  $-24$  or  $-21$

26  $-14$  or  $3$

27  $-30$  or  $-28$

28  $-30$  or  $-17$

## Medium

Questions 29–37

Write the integers in ascending order (smallest first).

**WE DO** With your teacher: copy the working.

E1  $15, -8, -22, -9$

E2  $5, 14, -24, -18$

**YOU DO** Your turn.

29  $5, -20, 3, 0$

smallest:  $-20$ , largest:  $5$

$-20, 0, 3, 5$

30  $-24, 5, 15, -6$

smallest:  $-24$ , largest:  $15$

31  $10, -7, -8, 4$

smallest: \_\_\_\_\_, largest: \_\_\_\_\_

32  $-23, -14, 4, 15$

33  $6, -16, 20, -25$

34  $-10, -19, 17, 9$

Write the integers in descending order (largest first).

35  $12, -27, 25, -32$

36  $9, -22, -16, -13$

37  $-39, -23, -31, -2$

## Challenging

Questions 38–45

Simplify any  $-(-n)$  first, then answer the question.

**WE DO** With your teacher: copy the working.

E1 Write three integers greater than  $-14$  but less than  $-9$ .

E2 Which is the smallest:  $-(-7)$ ,  $-7$  or  $-(-12)$ ?

**YOU DO** Your turn.

38 Write in ascending order:  $-(-2)$ ,  $-9$ ,  $-12$ ,  $0$

$-(-2) = 2$

39 Write three integers greater than  $-18$  but less than  $-12$ .

integers from \_\_\_\_\_ to \_\_\_\_\_

e.g. \_\_\_\_\_, \_\_\_\_\_, \_\_\_\_\_

40 Which is the smallest:  $-(-7)$ ,  $-8$  or  $-(-9)$ ?

41 Write in ascending order:  $-(-4)$ ,  $-7$ ,  $-3$ ,  $0$

42 Write three integers greater than  $-19$  but less than  $-11$ .

43 Which is the smallest:  $-(-2)$ ,  $-6$  or  $-(-11)$ ?

44 Write in ascending order:  $-(-9)$ ,  $-7$ ,  $-11$ ,  $0$

45 Write three integers greater than  $-17$  but less than  $-12$ .

## Easy

Questions 1–28

Evaluate each sum. Start at the first number and move along the number line.



**WE DO** With your teacher: copy the working.

E1  $-4 + 8$  \_\_\_\_\_

E2  $-7 + 10$  \_\_\_\_\_

**YOU DO** Your turn.

1  $-7 + 1$          -6        

2  $-4 + 10$  \_\_\_\_\_

3  $-4 + 3$  \_\_\_\_\_

4  $-3 + 1$  \_\_\_\_\_

5  $-11 + 2$  \_\_\_\_\_

6  $-2 + 1$  \_\_\_\_\_

7  $-8 + 10$  \_\_\_\_\_

8  $-8 + 2$  \_\_\_\_\_

9  $-10 + 11$  \_\_\_\_\_

10  $-9 + 5$  \_\_\_\_\_

11  $-7 + 7$  \_\_\_\_\_

12  $-11 + 11$  \_\_\_\_\_

13  $-10 + 7$  \_\_\_\_\_

14  $-9 + 8$  \_\_\_\_\_

15  $-5 + 5$  \_\_\_\_\_

16  $-2 + 8$  \_\_\_\_\_

17  $-1 + 12$  \_\_\_\_\_

18  $-8 + 12$  \_\_\_\_\_

Evaluate. Adding a negative is the same as subtracting.



19  $8 + (-1)$  \_\_\_\_\_

20  $11 + (-1)$  \_\_\_\_\_

21  $-6 + (-7)$  \_\_\_\_\_

22  $10 + (-11)$  \_\_\_\_\_

23  $-4 + (-12)$  \_\_\_\_\_

24  $11 + (-4)$  \_\_\_\_\_

25  $12 + (-2)$  \_\_\_\_\_

26  $12 + (-3)$  \_\_\_\_\_

27  $7 + (-1)$  \_\_\_\_\_

28  $2 + (-11)$  \_\_\_\_\_

## Medium

Questions 29–38

Evaluate. Add from left to right.

**WE DO** With your teacher: copy the working.

E1  $-12 + (-11) + (-13)$  \_\_\_\_\_

E2  $-4 + (-3) + (-6)$  \_\_\_\_\_

**YOU DO** Your turn.

29  $-4 + 14 + (-5)$

$= 10 + (-5)$

$= 5$

30  $-7 + (-3) + 9$

$= -10 + 9$

$=$  \_\_\_\_\_

31  $10 + 2 + 5$

$=$  \_\_\_\_\_ + \_\_\_\_\_

$=$  \_\_\_\_\_

32  $-15 + 4 + 1$  \_\_\_\_\_

33  $3 + (-8) + (-5)$  \_\_\_\_\_

34  $-13 + 7 + 4$  \_\_\_\_\_

35  $13 + 9 + 2$  \_\_\_\_\_

36  $6 + 4 + (-3)$  \_\_\_\_\_

37  $4 + 6 + (-7)$  \_\_\_\_\_

38  $-9 + 14 + 1$  \_\_\_\_\_

## Challenging

Questions 39–45

Write a number sentence, then answer the question.



**WE DO** With your teacher: copy the working.

E1 The temperature is  $-7^{\circ}\text{C}$  and rises  $8^{\circ}\text{C}$ . What is the new temperature?

E2 A lift at level  $-3$  goes up 6 floors. Which level is it on?

**YOU DO** Your turn.

39 Sam has \$35 and spends \$51 on credit. What is the balance?

$35 + (-51)$

$=$  \_\_\_\_\_

40 Find the missing number:  $-12 + \square = 6$

from \_\_\_\_\_ to \_\_\_\_\_ is \_\_\_\_\_ to the right

$\square =$  \_\_\_\_\_

41 The temperature is  $-6^{\circ}\text{C}$  and rises  $16^{\circ}\text{C}$ . What is the new temperature?

42 A lift at level  $-2$  goes up 5 floors. Which level is it on?

43 Sam has \$34 and spends \$62 on credit. What is the balance?

44 Find the missing number:  $-5 + \square = 2$

45 The temperature is  $-12^{\circ}\text{C}$  and rises  $6^{\circ}\text{C}$ . What is the new temperature?

## Easy

Questions 1–28

Evaluate. Subtracting moves you left on the number line.



**WE DO** With your teacher: copy the working.

E1  $-6 - 4$  \_\_\_\_\_

E2  $-5 - 4$  \_\_\_\_\_

**YOU DO** Your turn.

1  $-8 - 7$  -15

2  $-8 - 15$  \_\_\_\_\_

3  $2 - 2$  \_\_\_\_\_

4  $5 - 15$  \_\_\_\_\_

5  $5 - 13$  \_\_\_\_\_

6  $9 - 12$  \_\_\_\_\_

7  $6 - 10$  \_\_\_\_\_

8  $1 - 3$  \_\_\_\_\_

9  $-3 - 3$  \_\_\_\_\_

10  $0 - 8$  \_\_\_\_\_

11  $0 - 10$  \_\_\_\_\_

12  $8 - 2$  \_\_\_\_\_

13  $-8 - 10$  \_\_\_\_\_

14  $-6 - 3$  \_\_\_\_\_

15  $-1 - 9$  \_\_\_\_\_

16  $6 - 13$  \_\_\_\_\_

17  $4 - 3$  \_\_\_\_\_

18  $-5 - 10$  \_\_\_\_\_

Evaluate. Subtracting a negative is the same as adding.



19  $12 - (-9)$  \_\_\_\_\_

20  $-2 - (-11)$  \_\_\_\_\_

21  $10 - (-1)$  \_\_\_\_\_

22  $8 - (-8)$  \_\_\_\_\_

23  $2 - (-9)$  \_\_\_\_\_

24  $-12 - (-5)$  \_\_\_\_\_

25  $3 - (-3)$  \_\_\_\_\_

26  $11 - (-10)$  \_\_\_\_\_

27  $-4 - (-8)$  \_\_\_\_\_

28  $-6 - (-1)$  \_\_\_\_\_

## Medium

Questions 29–35

Evaluate. Change each  $- (-)$  to  $+$ , then add from left to right.

**WE DO** With your teacher: copy the working.

E1  $1 - (-15) + (-15)$  \_\_\_\_\_

E2  $7 - (-13) + (-6)$  \_\_\_\_\_

**YOU DO** Your turn.

29  $-13 - (-7) + (-7)$

$= -13 + 7 + (-7)$

$= -6 + (-7)$

$= -13$

30  $-5 - (-1) + 15$

$= -5 + 1 + 15$

$= -4 + 15$

$= 11$

31  $14 - (-10) + 11$

$= 14 + 10 + 11$

$= 35$

$= 35$

32  $13 - (-7) + (-5)$

33  $6 - (-9) + (-5)$

34  $9 - (-7) + (-7)$

35  $6 - (-6) + (-2)$

## Challenging

Questions 36–41

Write a number sentence, then answer the question.



**WE DO** With your teacher: copy the working.

E1 The minimum temperature was  $-5^{\circ}\text{C}$  and the maximum was  $19^{\circ}\text{C}$ . By how much did it rise?

E2 A submarine at  $-62$  m rises 22 m. Where is it now?

**YOU DO** Your turn.

36 What is 2 less than  $-9$ ?

$-9 - 2$

$= -11$

37 Find the difference between  $-6$  m and 24 m.

$24 - (-6)$

$= 24 + 6$

$= 30$  m

38 The minimum temperature was  $-10^{\circ}\text{C}$  and the maximum was  $11^{\circ}\text{C}$ . By how much did it rise?

39 A submarine at  $-105$  m rises 22 m. Where is it now?

40 What is 9 less than  $-4$ ?

41 Find the difference between  $-15$  m and 18 m.

## Easy

Questions 1–28

Evaluate each product.  
Same signs: positive.  
Different signs: negative.

**WE DO** With your teacher: copy the working.

E1  $-5 \times 3$  \_\_\_\_\_

E2  $6 \times (-2)$  \_\_\_\_\_

**YOU DO** Your turn.

1  $-7 \times (-9)$  \_\_\_\_\_

63

2  $-7 \times 7$  \_\_\_\_\_

3  $-3 \times (-1)$  \_\_\_\_\_

4  $1 \times (-3)$  \_\_\_\_\_

5  $-1 \times (-7)$  \_\_\_\_\_

6  $8 \times (-9)$  \_\_\_\_\_

7  $4 \times (-9)$  \_\_\_\_\_

8  $-8 \times 8$  \_\_\_\_\_

9  $-10 \times 8$  \_\_\_\_\_

10  $5 \times (-1)$  \_\_\_\_\_

11  $-7 \times (-8)$  \_\_\_\_\_

12  $-4 \times 9$  \_\_\_\_\_

13  $-8 \times (-5)$  \_\_\_\_\_

14  $-8 \times 7$  \_\_\_\_\_

15  $-3 \times (-10)$  \_\_\_\_\_

16  $8 \times (-3)$  \_\_\_\_\_

17  $6 \times (-5)$  \_\_\_\_\_

18  $-2 \times (-2)$  \_\_\_\_\_

Write the sign of the answer only: + or -.

19  $2 \times (-7)$  \_\_\_\_\_

20  $(-1) \times 9$  \_\_\_\_\_

21  $3 \times 1$  \_\_\_\_\_

22  $2 \times 5$  \_\_\_\_\_

23  $(-2) \times 1$  \_\_\_\_\_

24  $(-3) \times 3 \times 4$  \_\_\_\_\_

25  $7 \times 7$  \_\_\_\_\_

26  $4 \times (-3)$  \_\_\_\_\_

27  $1 \times (-1) \times 3$  \_\_\_\_\_

28  $9 \times (-6) \times 6$  \_\_\_\_\_

## Medium

Questions 29–48

Evaluate. Multiply two numbers at a time.

**WE DO** With your teacher: copy the working.

E1  $4 \times (-5) \times (-3)$  \_\_\_\_\_

E2  $1 \times (-1) \times 4$  \_\_\_\_\_

**YOU DO** Your turn.

29  $-6 \times (-2) \times (-3)$  \_\_\_\_\_

$= 12 \times (-3)$

$= -36$

31  $-6 \times 5 \times (-5)$  \_\_\_\_\_

$= \_\_\_\_ \times (\_\_\_\_)$

$= \_\_\_\_$

33  $-3 \times 4 \times 5$  \_\_\_\_\_

35  $-3 \times 3 \times 2$  \_\_\_\_\_

37  $5 \times (-5) \times 4$  \_\_\_\_\_

39  $5 \times 4 \times (-5)$  \_\_\_\_\_

41  $-4 \times (-1) \times 4$  \_\_\_\_\_

43  $-4 \times 6 \times (-1)$  \_\_\_\_\_

45  $2 \times (-4) \times 1$  \_\_\_\_\_

47  $-4 \times 1 \times 4$  \_\_\_\_\_

30  $5 \times (-1) \times (-2)$  \_\_\_\_\_

$= -5 \times (-2)$

$= \_\_\_\_$

32  $-1 \times (-1) \times (-1)$  \_\_\_\_\_

34  $3 \times (-4) \times 1$  \_\_\_\_\_

36  $-1 \times 1 \times 2$  \_\_\_\_\_

38  $-3 \times 6 \times (-1)$  \_\_\_\_\_

40  $5 \times (-4) \times 1$  \_\_\_\_\_

42  $-5 \times 1 \times (-1)$  \_\_\_\_\_

44  $-1 \times (-5) \times (-3)$  \_\_\_\_\_

46  $-6 \times 5 \times (-4)$  \_\_\_\_\_

48  $-6 \times (-2) \times 2$  \_\_\_\_\_

## Challenging

Questions 49–55

Write a number sentence, then answer the question.

**WE DO** With your teacher: copy the working.

E1 A diver goes down 2 m every minute. Where is she after 3 minutes?

E2 The temperature drops  $4^\circ\text{C}$  each hour for 3 hours. What is the total change?

**YOU DO** Your turn.

49 Find the missing number:  $-4 \times \square = -12$

$\square = -12 \div (-4)$

$\square = \_\_\_\_$

50 A diver goes down 2 m every minute. Where is she after 6 minutes?

$\_\_\_\_ \times \_\_\_\_$

$= \_\_\_\_ \text{ m}$

51 The temperature drops  $6^\circ\text{C}$  each hour for 6 hours. What is the total change?

52 Find the missing number:  $-4 \times \square = 32$

53 A diver goes down 4 m every minute. Where is she after 7 minutes?

54 The temperature drops  $5^\circ\text{C}$  each hour for 9 hours. What is the total change?

55 Find the missing number:  $-4 \times \square = -8$



### Easy

Questions 1–28

Evaluate each quotient.  
Same signs: positive.  
Different signs: negative.

**WE DO** With your teacher: copy the working.

E1  $36 \div (-4)$

E2  $-28 \div (-7)$

**YOU DO** Your turn.

1  $8 \div (-8)$

-1

2  $90 \div (-9)$

3  $-4 \div (-1)$

4  $40 \div (-8)$

5  $4 \div (-1)$

6  $-40 \div (-5)$

7  $-56 \div 7$

8  $40 \div (-5)$

9  $-30 \div (-10)$

10  $-12 \div 3$

11  $-4 \div (-4)$

12  $-10 \div 1$

13  $-24 \div 6$

14  $15 \div (-5)$

15  $15 \div (-3)$

16  $-4 \div 4$

17  $-90 \div 9$

18  $30 \div (-3)$

Find the missing number.

19  $30 \div \square = -5$

20  $36 \div \square = 4$

21  $8 \div \square = 8$

22  $25 \div \square = 5$

23  $-14 \div \square = 7$

24  $-5 \div \square = 5$

25  $-5 \div \square = -5$

26  $-9 \div \square = 9$

27  $-20 \div \square = 4$

28  $-21 \div \square = 7$

### Medium

Questions 29–48

Evaluate. Divide from left to right.

**WE DO** With your teacher: copy the working.

E1  $8 \div 4 \div 2$

E2  $-40 \div (-5) \div (-2)$

**YOU DO** Your turn.

29  $-10 \div 1 \div 2$

$= -10 \div 2$

$= -5$

31  $8 \div (-1) \div (-4)$

$= \_\_\_\_ \div (\_\_\_\_)$

$= \_\_\_\_$

33  $-8 \div 2 \div (-2)$

30  $-9 \div 1 \div 3$

$= -9 \div 3$

$= \_\_\_\_$

32  $-8 \div 2 \div 4$

34  $-15 \div (-5) \div 3$

35  $-50 \div (-5) \div (-2)$

36  $-32 \div (-4) \div (-4)$

37  $45 \div 5 \div 3$

38  $10 \div 1 \div (-2)$

39  $4 \div 4 \div 1$

40  $32 \div (-4) \div 4$

41  $16 \div 4 \div 2$

42  $-60 \div 3 \div (-4)$

43  $36 \div 3 \div 4$

44  $6 \div (-2) \div 1$

45  $24 \div (-3) \div (-4)$

46  $-6 \div (-1) \div 3$

47  $-9 \div 3 \div 1$

48  $50 \div 5 \div (-2)$

### Challenging

Questions 49–55

Write a number sentence, then answer the question.

**WE DO** With your teacher: copy the working.

E1 A debt of \$160 is shared equally by 8 people. Write each share as an integer.

E2 The temperature fell  $15^\circ\text{C}$  over 3 hours. What was the change each hour?

**YOU DO** Your turn.

49 Find the missing number:  $\square \div 8 = 4$

$\square = 4 \times 8$

$\square = \_\_\_\_$

50 A debt of \$114 is shared equally by 6 people. Write each share as an integer.

$\_\_\_\_ \div \_\_\_\_$

$= \_\_\_\_ \text{ each}$

51 The temperature fell  $12^\circ\text{C}$  over 6 hours. What was the change each hour?

52 Find the missing number:  $\square \div (-4) = -5$

53 A debt of \$126 is shared equally by 7 people. Write each share as an integer.

54 The temperature fell  $24^\circ\text{C}$  over 4 hours. What was the change each hour?

55 Find the missing number:  $\square \div (-9) = 1$

## Easy

Questions 1–28

Evaluate.  $\times$  and  $\div$  before  $+$  and  $-$ .



**WE DO** With your teacher: copy the working.

E1  $3 + (-6) \times (-6)$  \_\_\_\_\_

E2  $-9 - 5 \times (-6)$  \_\_\_\_\_

**YOU DO** Your turn.

1  $-5 - (-6) \times (-4)$  \_\_\_\_\_

**-29**

2  $-12 - (-2) \times (-2)$  \_\_\_\_\_

3  $-6 - (-5) \times (-6)$  \_\_\_\_\_

4  $10 + 6 \times 4$  \_\_\_\_\_

5  $2 + 4 \times 5$  \_\_\_\_\_

6  $-3 - (-4) \times 4$  \_\_\_\_\_

7  $7 + (-6) \times (-2)$  \_\_\_\_\_

8  $9 - 2 \times (-2)$  \_\_\_\_\_

9  $3 - (-4) \times 4$  \_\_\_\_\_

10  $9 + 4 \times 2$  \_\_\_\_\_

11  $8 + 5 \times (-6)$  \_\_\_\_\_

12  $-10 + 2 \times (-6)$  \_\_\_\_\_

13  $-4 - 2 \times (-6)$  \_\_\_\_\_

14  $8 + (-1) \times (-3)$  \_\_\_\_\_

15  $1 + 3 \times (-1)$  \_\_\_\_\_

16  $-2 - 5 \times 5$  \_\_\_\_\_

17  $-6 - 4 \times 2$  \_\_\_\_\_

18  $5 - (-1) \times 5$  \_\_\_\_\_

Evaluate. Brackets first.



19  $(-6 - (-1)) \times 2$  \_\_\_\_\_

20  $(2 + (-9)) \times 3$  \_\_\_\_\_

21  $(-5 - 7) \times 4$  \_\_\_\_\_

22  $(6 - 3) \times 3$  \_\_\_\_\_

23  $(2 + (-4)) \times 1$  \_\_\_\_\_

24  $(4 - (-9)) \times 1$  \_\_\_\_\_

25  $(7 - (-1)) \times (-4)$  \_\_\_\_\_

26  $(-5 - (-6)) \times 5$  \_\_\_\_\_

27  $(-6 + 4) \times (-2)$  \_\_\_\_\_

28  $(-8 - 1) \times 5$  \_\_\_\_\_

## Medium

Questions 29–38

Evaluate. Do  $\times$  and  $\div$  first (left to right), then  $+$  and  $-$ .

**WE DO** With your teacher: copy the working.

E1  $6 \times 1 + 2 \div 1$  \_\_\_\_\_

E2  $3 \times (-6) + (-16) \div 4$  \_\_\_\_\_

**YOU DO** Your turn.

29  $-5 \times (-6) - (-5) \div (-5)$  \_\_\_\_\_

**= 30 - 1**

**= 29**

30  $-1 \times (-2) + (-10) \div 2$  \_\_\_\_\_

**= 2 + (-5)**

**= -3**

31  $1 \times 6 + (-10) \div 2$  \_\_\_\_\_

**= 6 + (-5)**

**= 1**

32  $1 \times 3 + 6 \div (-3)$  \_\_\_\_\_

33  $4 \times 1 + 15 \div 5$  \_\_\_\_\_

34  $6 \times (-4) - 8 \div 4$  \_\_\_\_\_

35  $6 \times (-2) - 20 \div 4$  \_\_\_\_\_

36  $-5 \times (-3) + 5 \div (-5)$  \_\_\_\_\_

37  $-6 \times 6 - 6 \div (-3)$  \_\_\_\_\_

38  $3 \times (-4) - 16 \div (-4)$  \_\_\_\_\_

## Challenging

Questions 39–45

Evaluate. Brackets first, then  $\times$  and  $\div$ , then  $+$  and  $-$ . One step on each line.



**WE DO** With your teacher: copy the working.

E1  $7 - (-4 + 6) \times (-3)$  \_\_\_\_\_

E2  $6 - (-1 + (-2)) \times (-1)$  \_\_\_\_\_

**YOU DO** Your turn.

39  $-5 - (7 + 6) \times (-3)$  \_\_\_\_\_

**= -5 - 13 \times (-3)**

**= -5 - (-39)**

**= 34**

40  $5 - (2 + 8) \times (-1)$  \_\_\_\_\_

**= 5 - 10 \times (-1)**

**= 5 - (-10)**

**= 15**

41  $-5 - (5 + 8) \times (-3)$  \_\_\_\_\_

42  $2 - (7 + (-4)) \times (-4)$  \_\_\_\_\_

43  $7 - (8 + (-1)) \times 3$  \_\_\_\_\_

44  $-7 - (-4 + 5) \times (-3)$  \_\_\_\_\_

45  $4 - (7 + 7) \times (-4)$  \_\_\_\_\_

## Easy

Questions 1–28

Use your calculator. Use the  $(-)$  key for negative numbers.



**WE DO** With your teacher: copy the working.

**E1**  $252 + (-391)$  \_\_\_\_\_

**E2**  $-369 + 350$  \_\_\_\_\_

**YOU DO** Your turn.

**1**  $27 + 17$  \_\_\_\_\_ **44**

**2**  $-160 + (-124)$  \_\_\_\_\_

**3**  $24 + 106$  \_\_\_\_\_

**4**  $-198 + 77$  \_\_\_\_\_

**5**  $-269 + 255$  \_\_\_\_\_

**6**  $269 + (-274)$  \_\_\_\_\_

**7**  $267 + 146$  \_\_\_\_\_

**8**  $-158 + 224$  \_\_\_\_\_

**9**  $-396 + 241$  \_\_\_\_\_

**10**  $197 + (-106)$  \_\_\_\_\_

**11**  $59 + (-266)$  \_\_\_\_\_

**12**  $-303 + (-27)$  \_\_\_\_\_

**13**  $296 + 12$  \_\_\_\_\_

**14**  $224 + 100$  \_\_\_\_\_

**15**  $127 + 343$  \_\_\_\_\_

**16**  $-299 + 318$  \_\_\_\_\_

**17**  $254 + 3$  \_\_\_\_\_

**18**  $-72 + 281$  \_\_\_\_\_

Use your calculator.



**19**  $43 \times (-4)$  \_\_\_\_\_

**20**  $22 \times 3$  \_\_\_\_\_

**21**  $45 \times (-44)$  \_\_\_\_\_

**22**  $3 \times (-11)$  \_\_\_\_\_

**23**  $-45 \times (-20)$  \_\_\_\_\_

**24**  $-22 \times 12$  \_\_\_\_\_

**25**  $17 \times (-33)$  \_\_\_\_\_

**26**  $31 \times (-32)$  \_\_\_\_\_

**27**  $13 \times (-33)$  \_\_\_\_\_

**28**  $33 \times (-13)$  \_\_\_\_\_

## Medium

Questions 29–38

Use your calculator. Enter the brackets, then check the answer makes sense.



**WE DO** With your teacher: copy the working.

**E1**  $(-86 - (-81)) \times (-16)$  \_\_\_\_\_

**E2**  $(-3 - 67) \times (-11)$  \_\_\_\_\_

**YOU DO** Your turn.

**29**  $(-86 - (-8)) \times (-22)$  \_\_\_\_\_

**bracket: -78**

$-78 \times (-22) = 1\ 716$

**30**  $(12 - 22) \times 19$  \_\_\_\_\_

**bracket: -10**

$_____ \times _____ = _____$

**31**  $(12 - (-59)) \times (-18)$  \_\_\_\_\_

**bracket: \_\_\_\_\_**

$_____ \times (_____) = _____$

**32**  $(-44 - 62) \times (-21)$  \_\_\_\_\_

**33**  $(45 - 80) \times (-4)$  \_\_\_\_\_

**34**  $(-90 - 59) \times (-19)$  \_\_\_\_\_

**35**  $(-31 - 52) \times (-15)$  \_\_\_\_\_

**36**  $(57 - 83) \times 15$  \_\_\_\_\_

**37**  $(-74 - (-90)) \times (-3)$  \_\_\_\_\_

**38**  $(-62 - 84) \times (-15)$  \_\_\_\_\_

## Challenging

Questions 39–46

Estimate first, then use your calculator. Is your answer close to the estimate?



**WE DO** With your teacher: copy the working.

**E1** Estimate  $-60 \times 44$ , then calculate.

**E2** Estimate  $-32 \times 48$ , then calculate.

**YOU DO** Your turn.

**39** Estimate  $-34 \times 17$ , then calculate.

**estimate:**  $-30 \times 20 \approx -600$

**calculator:** \_\_\_\_\_

**40** Estimate  $-58 \times 36$ , then calculate.

**estimate:**  $_____ \times _____ \approx _____$

**calculator:** \_\_\_\_\_

**41** Estimate  $-42 \times 35$ , then calculate.

**42** Estimate  $-46 \times 28$ , then calculate.

**43** Estimate  $-38 \times 30$ , then calculate.

**44** Estimate  $-57 \times 13$ , then calculate.

**45** Estimate  $-48 \times 26$ , then calculate.

**46** Estimate  $-85 \times 12$ , then calculate.

### Easy

Questions 1–28

Find the new temperature.

**WE DO** With your teacher: copy the working.

**E1** 5°C, rises 14°C

**E2** 14°C, falls 4°C

**YOU DO** Your turn.

**1** -4°C, falls 17°C

-21°C

**2** -7°C, falls 16°C

**3** -9°C, rises 15°C

**4** 11°C, falls 7°C

**5** 7°C, falls 5°C

**6** -10°C, rises 10°C

**7** 5°C, falls 11°C

**8** -2°C, falls 15°C

**9** 11°C, rises 10°C

**10** 12°C, falls 13°C

**11** -2°C, falls 12°C

**12** 13°C, falls 17°C

**13** -10°C, falls 14°C

**14** 13°C, rises 13°C

**15** 8°C, rises 4°C

**16** -6°C, falls 3°C

**17** 14°C, rises 15°C

**18** -9°C, rises 12°C

Find the new bank balance.

**19** -\$72, deposit \$92

**20** -\$9, withdraw \$103

**21** -\$23, deposit \$24

**22** \$133, deposit \$70

**23** \$118, deposit \$53

**24** -\$73, withdraw \$18

**25** \$103, withdraw \$78

**26** -\$9, deposit \$86

**27** \$13, deposit \$42

**28** -\$21, withdraw \$64

### Medium

Questions 29–36

Find the difference between the two heights. Write a number sentence.

**WE DO** With your teacher: copy the working.

**E1** a hill 267 m above sea level and a reef 57 m below

**E2** a hill 280 m above sea level and a reef 74 m below

**YOU DO** Your turn.

**29** a hill 106 m above sea level and a reef 10 m below

106 - (-10)

= 116 m

**30** a hill 66 m above sea level and a reef 19 m below

66 - (-19)

= \_\_\_\_ m

**31** a hill 52 m above sea level and a reef 32 m below

\_\_\_\_ - (\_\_\_\_)

= \_\_\_\_ m

**32** a hill 174 m above sea level and a reef 48 m below

**33** a hill 388 m above sea level and a reef 57 m below

**34** a hill 129 m above sea level and a reef 45 m below

**35** a hill 28 m above sea level and a reef 88 m below

**36** a hill 247 m above sea level and a reef 33 m below

### Challenging

Questions 37–41

Write a number sentence for each step, then answer the question.

**WE DO** With your teacher: copy the working.

**E1** A diver is at -45 m. She rises 4 m a minute for 4 minutes. Where is she now?

**E2** A game scores +4 for a win and -3 for a loss. Mia wins 3 and loses 5. What is her score?

**YOU DO** Your turn.

**37** It is -2°C at 6 a.m. It warms 3°C an hour for 3 hours. What is the temperature?

 $3 \times 3 = 9$ 

-2 + 9

= \_\_\_\_ °C

**38** A diver is at -30 m. She rises 5 m a minute for 5 minutes. Where is she now?

\_\_\_\_ × \_\_\_\_ = \_\_\_\_

\_\_\_\_ + \_\_\_\_

= \_\_\_\_ m

**39** A game scores +4 for a win and -1 for a loss. Mia wins 4 and loses 4. What is her score?

**40** It is -3°C at 6 a.m. It warms 5°C an hour for 6 hours. What is the temperature?

**41** A diver is at -45 m. She rises 3 m a minute for 2 minutes. Where is she now?